



MATER DEI
COLLEGE
FAITH WITH COURAGE

Semester Two Examination, 2022

Question/Answer booklet

MATHEMATICS APPLICATIONS UNITS 3&4

Section Two: Calculator-assumed

If required by your examination administrator, please
place your student identification label in this box

WA student number: In figures

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In words

Your name

Solly

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper,
and up to three calculators, which can include scientific, graphic and
Computer Algebra System (CAS) calculators, are permitted in this ATAR
course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that
you do not have any unauthorised material. If you have any unauthorised material with you, hand
it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	50	35
Section Two: Calculator-assumed	12	12	100	99	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed

65% (99 Marks)

This section has **twelve** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8

(6 marks)

The number of tumble driers sold each month at a store are shown in the table below.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Number	44	39	46	56	56	65	74	81	69	—

(a) Determine

(i) the three-point moving average for March.

(1 mark)

$$(39 + 46 + 56) \div 3 = 47$$

✓ correct value

(ii) the six-point moving average centred on May.

(2 marks)

$$\left(\frac{39}{2} + 46 + 56 + 56 + 65 + 74 + \frac{81}{2} \right) \div 6 = 59.5$$

✓ indicates correct calculation

✓ evaluates correctly

(b) Explain the purpose of centring a moving average with an even number of data points.

(1 mark)

To align the moving average with time (the middle of the time period)

✓ reasonable explanation

(c) Determine the number of tumble driers sold in October, given that the four-point moving average for the data, centred on August, is 71.

(2 marks)

$$\left(\frac{65}{2} + 74 + 81 + 69 + \frac{x}{2} \right) \div 4 = 71$$

$$x = 55$$

Number sold was 55

✓ indicates suitable method
✓ states correct value

Question 9

(6 marks)

An injection moulding machine was purchased by a business to make plastic chairs. The initial value of the machine was \$7500, and this value depreciates at a constant rate of 8 cents per chair made.

- (a) Calculate the loss in value of the machine after 2000 chairs have been made. (1 mark)

$$2000 \times 0.08 \\ = \$160$$

Machine has lost \$160 in value

✓ correct loss in value

The value of the machine, in dollars, after n chairs have been made is given by $T_n = a - bn$.

- (b) State the value of the constant a and the value of the constant b . (2 marks)

$$T_n = 7500 - 0.08n$$

Hence $a = 7500$ and $b = 0.08$

✓ value of a
✓ value of b

- (c) Calculate the value of the machine after 33 550 chairs have been made. (1 mark)

$$T_{33550} = \$4816$$

✓ correct value

- (d) The machine will be scrapped once its value falls to \$700. Determine the number of chairs that the machine must make to reach this value. (2 marks)

$$7500 - 0.08n = 700$$

$$n = 85000$$

It must produce 85000 chairs

✓ correctly forms equation
✓ correct number of chairs

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 10

(6 marks)

During the spring of 2020, a national real estate company observed that from a sample of 320 households in Perth who rented their accommodation and applied to their landlord for a rent reduction, 192 applications were approved. From a similar survey in Melbourne of 400 households, 236 applications for a rent reduction were approved.

- (a) Use the above information to complete the two-way frequency table below. (2 marks)

Rent Reduction / City	Perth	Melbourne
Approved	192	236
Not approved	128	164
Total	320	400

✓ one correct column
✓ correct table

The real estate company wanted to know whether the data provided any clear evidence of the presence of an association between rent reduction approvals and city.

- (b) Complete the column percentaged two-way frequency table below. (2 marks)

Percentages	Perth	Melbourne
Approved	60	59
Not approved	40	41
Total	100	100

✓ one correct column
✓ correct table

- (c) State, with justification, whether the data provides clear evidence of the presence of an association between rent reduction approvals and city. (2 marks)

There is no clear evidence of an association, as the percentage approval rates for both cities are close to each other.

✓ state data does not provide clear evidence

✓ justifies by referring to similar percentage along rows.

Question 11

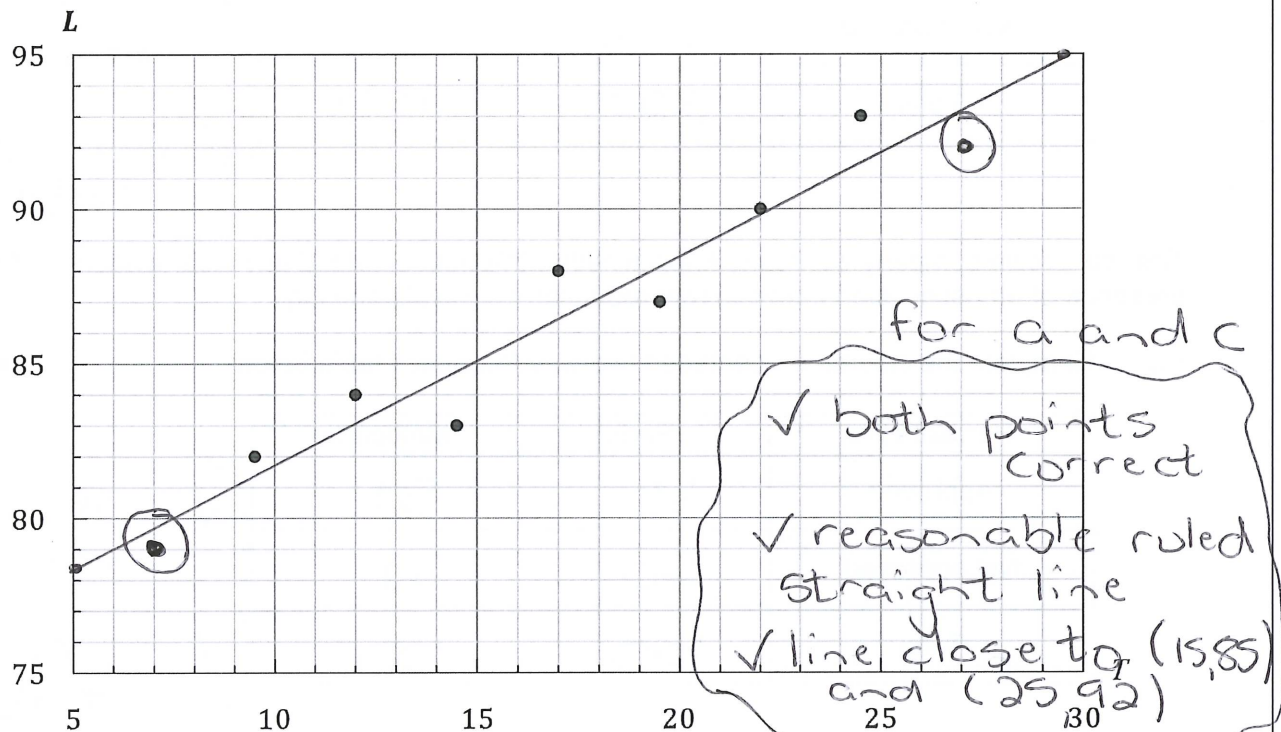
(12 marks)

A researcher is analysing data to confirm Allen's Rule, a century-old biological observation that an association exists between ambient temperature and tail length in mammals. The table shows the tail length in millimetres of 6-week-old mice and the temperature in degrees Celsius that they were housed at since birth.

Temperature, T	7.0	9.5	12.0	14.5	17.0	19.5	22.0	24.5	27.0
Tail length, L	79	82	84	83	88	87	90	93	92

- (a) On the scatterplot below, plot and circle the two missing data points from the table.

(1 mark)



- (b) Determine the equation of the least-squares line for the data.

(2 marks)

$$\hat{L} = 0.673T + 74.998$$

(to at least 2sf)
 ✓ correctly determines coefficients
 ✓ correctly writes equation using given variables

- (c) Draw the least-squares line on the scatterplot above.

(2 marks)

- (d) Interpret the slope of the least-squares line in the context of this question.

(2 marks)

For every 1°C increase in temperature, the tail length increases by an average of 0.673 mm .

✓ refers to correct variables
 ✓ correctly states average length increase

- (e) Determine the value of the correlation coefficient between T and L , and use it to describe the association between the variables in terms of direction and strength. (2 marks)

$$r_{TL} = 0.966$$

Association is positive and strong.

✓ correct coefficient (shown to at least 2sf)
✓ correctly states direction + strength

- (f) Use the equation of the least-squares line to predict the tail length of a 6-week-old mouse that was housed at a temperature of 30°C , and comment on the validity of this prediction. (2 marks)

$$L = 0.673(30) + 74.998$$

$$= 95.2 \text{ mm.}$$

This prediction is not valid as it involves extrapolation.

✓ correct prediction (to at least 2sf)
✓ states not valid due to extrapolation

- (g) The researcher stated, "This data clearly shows that housing mice at higher temperatures causes their tails to grow longer". Comment on the validity of this statement. (1 mark)

The statement is not valid since an observed association between the variables does not necessarily mean a causal relationship exists.

✓ indicates statement not valid, with appropriate reason

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Question 12

(8 marks)

Bill is keen to buy a car and has been offered a reducing balance loan of \$22 100 to help with his purchase. The loan is to be repaid in 24 equal monthly payments of \$1010.85, and Bill has started the spreadsheet below to investigate how the balance of the loan reduces.

Month (n)	Opening balance	Interest	Repayment	Closing balance (T_n)
1	22 100.00	167.96	1010.85	21 257.11
2	21 257.11	161.55	1010.85	20 407.81
3	20 407.81			

- (a) Use figures from the first month to deduce that the annual interest rate is 9.12%. (1 mark)

$$167.96 \div 22\,100 \times 12 = 0.0076 \times 12 = 0.0912$$

Rate is 9.12% pa.

✓ Shows correct calculation(s) to obtain int rate

- (b) Determine the interest for the third month and hence state the closing balance for that month. (2 marks)

$$I = 20\,407.81 \times 0.0076 = \$155.10$$

$$T_3 = 20\,407.81 + 155.10 - 1010.85 = \$19\,552.06$$

✓ correctly calculates interest
✓ correct closing balance

- (c) Write a recurrence relation for T_n , the closing balance for month n . (2 marks)

$$T_{n+1} = 1.0076T_n - 1010.85, \quad T_0 = 22\,100$$

✓ States correct rule
✓ States correct initial term

- (d) Determine the closing balance of the loan at the end of the twelfth month. (1 mark)

$$T_{12} = \$11\,551.63$$

✓ correct closing balance

- (e) Determine the total interest paid on the loan over the 24 months. (2 marks)

$$\text{Repayments: } 24 \times 1010.85 = 24\,260.40$$

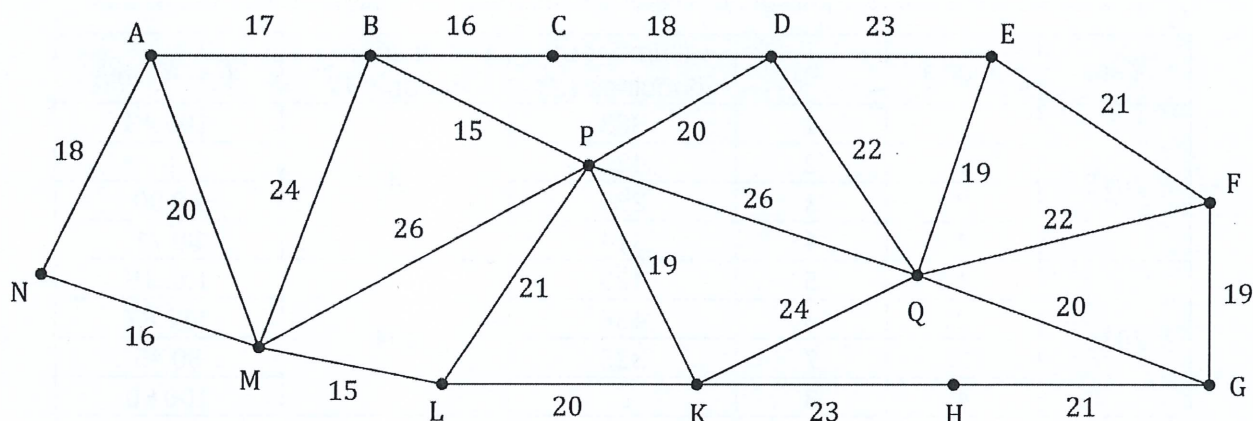
$$\text{Total Interest} = 24\,260.40 - 22\,100 = \$2\,160.40$$

✓ indicates correct method (possible financial calculator)
✓ correct total interest

Question 13

(7 marks)

The edges in the graph below represent power lines between 14 buildings, and the weight on each edge is the cost, in hundreds of dollars, to upgrade that line to carry more power.



- (a) Clearly show the minimum spanning tree on the graph and hence determine the cost of upgrading all the power lines that form the minimum spanning tree. (5 marks)

LMNABPK: $15 + 16 + 18 + 17 + 15 + 19 = 100$
BCDQE: $16 + 18 + 22 + 19 = 75$ QGH, GF: $21 + 20 + 19 = 60$

Hence sum of edges in tree is 235
and so cost is \$23500

- ✓ a spanning tree
- ✓ a spanning tree with at least 10 correct edges
- ✓ correct minimum spanning tree
- ✓ indicates correct sum of edges in tree
- ✓ correct cost of upgrade

- (b) Given that Prim's algorithm was used to determine the minimum spanning tree for the graph above, state the final edge that would be connected to complete the minimum spanning tree, when

- (i) building M was the first vertex used with the algorithm. (1 mark)
Vertices G and H

✓ correct pair of vertices

- (ii) building E was the first vertex used with the algorithm. (1 mark)
- Vertices P and K

✓ correct pair of vertices

Question 14

(11 marks)

The number of enquiries received each quarter by a party hire business are shown in the table below, together with some derived figures.

Year	Quarter	n	Number of enquiries (E)	Seasonal mean (M)	$E \div M$ (%)
2015	1	1	383	A	109.43
	2	2	423		B
	3	3	280		80.00
	4	4	314		89.71
2016	1	5	429	404	106.19
	2	6	456		112.87
	3	7	325		80.45
	4	8	C		100.50
2017	1	9	461	449	102.67
	2	10	509		113.36
	3	11	383		85.30
	4	12	443		98.66

- (a) Calculate the value of A , the value of B and the value of C in the table. (3 marks)

$$A = (383 + 423 + 280 + 314) \div 4 = 350$$

$$B = 423 \div 350 \times 100 = 120.86$$

$$C = 4 \times 404 - 429 - 456 - 325 \quad \text{or} \quad C = 404 \times 1.0050$$

$$= 406 \quad \quad \quad = 406$$

✓ value of A
 ✓ value of B
 ✓ value of C

- (b) Complete the table below, to show the seasonal index for each quarter. (2 marks)

Quarter	1	2	3	4
Seasonal index	1.061	1.157	0.819	0.963

$$Q3: (0.80 + 0.8045 + 0.853) \div 3 = 0.819$$

$$Q4: 4 - 1.061 - 1.157 - 0.819 = 0.963$$

✓ index for Q3
 ✓ index for Q4

- (c) Determine in which quarter of 2017 the deseasonalised number of enquiries was the highest, and state what this deseasonalised number is. (2 marks)

$$461 \div 1.061 = 434$$

$$509 \div 1.157 = 440$$

$$383 \div 0.819 = 468$$

$$443 \div 0.963 = 460$$

Highest in third quarter, when it was 468.

✓ states correct quarter
✓ states correct value

The equation of the least-squares line for the deseasonalised number of enquiries (e) against time period (n) is $\hat{e} = 11.51n + 326.33$, and the correlation coefficient is 0.91.

- (d) Use the equation of the least-square line above and seasonal adjustments as required to predict the number of enquiries received by the business in the second quarter of 2018. (3 marks)

$$\hat{e} = 11.51(14) + 326.33 = 487.47$$

$$487.42 \times 1.157 = 564.0$$

Predict 564 enquires.

✓ correctly uses $n=14$ in least sq. line
✓ correctly multiplies by seasonal index +
✓ rounds to a whole number

- (e) Time series predictions inevitably involve extrapolation. Ignoring this factor and assuming that a strong association exists, state an assumption required for a reasonable level of confidence in predictions such as that made in part (d). (1 marks)

One of

- existing trend in the number of enquires continues
- existing seasonality in the number of enquires continues.

✓ indicates trend or seasonality continues

Question 15

(7 marks)

The annual cost of electricity A_n for a household, in the n^{th} year after 2019, is shown in the table below in dollars.

Year	2019	2020	2021
n	0	1	2
A_n	1750.00	1890.00	2041.20

- (a) Use the values of A_n in the table to deduce that the annual cost of electricity increased by 8% every year. (2 marks)

$$1890 \div 1750 = 1.08$$

$$2041.20 \div 1890 = 1.08$$

Hence annual increase of 8%

✓ uses 2 consecutive terms to obtain r
 ✓ uses second pair of terms to confirm r

For the remainder of this question, assume that the annual cost of electricity for the household continues to increase by 8% each year.

- (b) Determine the annual cost of electricity for the household in 2022. (1 mark)

$$2041.20 \times 1.08 = \$2204.50$$

✓ correct cost, to nearest cent

- (c) Determine a rule for the n^{th} term of A_n . (2 marks)

$$A_n = 1750(1.08)^n$$

✓ uses correct coefficients in rule for geometric sequence

✓ correct rule

- (d) In which year will the annual cost of electricity for the household first exceed \$4000, and state the cost in this year. (2 marks)

$$A_{11} = 4080.37$$

When $n=11$, the year will be 2030
 and the cost will be \$4080.37

✓ states correct year
 ✓ states correct cost

Question 16

(7 marks)

Eight students were asked to complete a physical task using both hands and the time taken recorded as t_1 seconds. The students then repeated the task using just one hand, and the new time taken recorded as t_2 seconds. The table below shows the paired data for these students.

t_1	17	16	11	12	24	26	22	20
t_2	37	34	32	34	39	41	37	39

The equation of the least squares line for this data is $\hat{t}_2 = 0.52t_1 + 27.0$, and the correlation coefficient is 0.93.

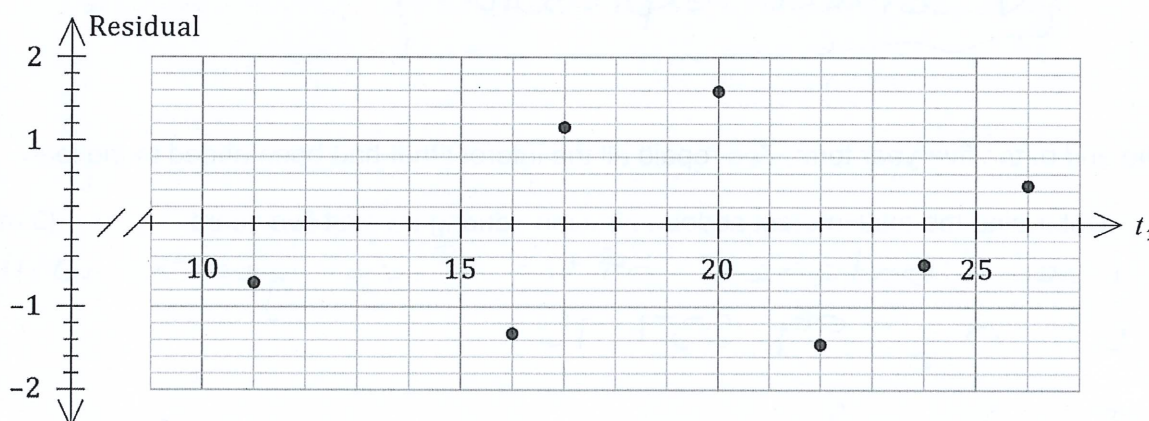
- (a) Which is the explanatory variable? (1 mark)

Time using both hands t_1
 (✓ states t_1)

- (b) What percentage of the variation in t_2 can be explained by the variation in t_1 ? (1 mark)

$r^2 = 0.93^2 = 0.8649$
 86% of the variation can be explained in this way.
 (✓ correct percentage (to at least 2sf))

The residual plot for seven of the eight data points is shown below.



- (c) Calculate the residual for the student with $t_1 = 12$ and add this point to the residual plot above. (3 marks)

$\hat{t}_2 = 0.52(12) + 27 = 33.24$
 $t_2 - \hat{t}_2 = 34 - 33.24 = 0.76$
 (✓ calculates $\hat{t}_2$
 ✓ calculates residual
 ✓ correctly plots point)

- (d) Use the residual plot to justify whether the least-squares line is an appropriate model for the relationship between t_1 and t_2 . (2 marks)

Is appropriate
 As no pattern is evident in the residual plot
 (✓ states model is appropriate
 ✓ justifies using no pattern evident in residual plot
 See next page)

Question 17

(12 marks)

Zoe arranged a loan of \$75 000 with her bank so that for the first five years, no repayments were required. The loan attracted interest of 5.64% per annum, compounded quarterly.

- (a) State a recurrence relation for A_n , the loan balance after n quarters. (2 marks)

$$0.0564 \div 4 = 0.0141$$

$$A_{n+1} = 1.0141 A_n \quad A_0 = 75\,000$$

✓ indicates correct multiplier

✓ states recurrence relation with initial term

- (b) The effective interest rate of the loan is 5.76% per annum. Write an expression that can be used to evaluate this rate from the advertised rate of 5.64% per annum. (1 mark)

$$\left(1 + \frac{0.0564}{4}\right)^4 - 1 \quad \text{or} \quad (1.0141)^4 - 1$$

✓ correct expression

At the end of the five-year term, Zoe repaid all the interest that had been added to the loan.

- (c) Determine the total interest added to the loan during the first five years. (3 marks)

Loan balance after five years will be $A_{20} = 99\,237.75$

$$\begin{aligned} I &= A_{20} - A_0 \\ &= 99\,237.75 - 75\,000 \\ &= 24\,237.75 \end{aligned}$$

Total interest is \$24 237.75

✓ indicates 20 compounding periods

✓ calculates loan balance

✓ correct total interest

After Zoe repaid the interest, the loan reverted to a reducing balance loan, with the principal of \$75 000 repaid in eight equal quarterly instalments. The first repayment was made three months after the end of the initial five-year term.

- (d) Determine the amount of each repayment.

(2 marks)

Using financial calculator with payment date set to End and

$$N=8, I\% = 5.64, PV = 75\ 000, FV = 0, C/Y = 4$$

$$P/Y = 4 \text{ then } PMT = -9979.56$$

Each repayment is \$9979.56

✓ indicates correct settings + values
✓ correctly solves for repayment

- (e) State the balance of the loan at the end of the sixth year, after four repayments have been made.

(2 marks)

Using financial calculator with payment date set to End and

$$N=4, I\% = 5.64, PV = 75\ 000, PMT = -9979.56$$

$$C/Y = 4, P/Y = 4 \text{ then } FV = -38\ 549.83$$

Balance will be \$38 549.83

✓ indicates correct setting + values
✓ correctly solves for balance

- (f) Determine the total interest added to the loan during the sixth year.

(2 marks)

Principal less 4 repayments is

$$75\ 000 - 4 \times 9979.56$$

$$= 35\ 081.76$$

$$\text{Total interest} = 38\ 549.83 - 35\ 081.76$$

$$= 3468.07$$

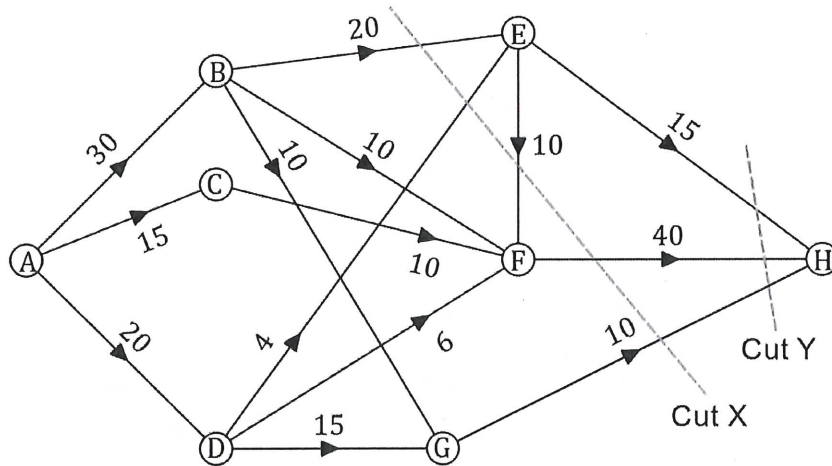
✓ indicates appropriate method (using financial calculator or other)

✓ correct total interest (2 marks for just correct answer OK)

Question 18

(9 marks)

The flow of air from intake A through a system of ducts to outlet H is shown in the network below. Each edge weight represents the maximum capacity of that duct, in cubic metres per minute.



(a) Determine the capacity of

(i) cut X. $= 20 + 10 + 10 + 40 + 10 = 74 \text{ m}^3/\text{min}$ (1 mark)

✓ correct capacity of cut X

(ii) cut Y. $= 15 + 40 + 10 = 65 \text{ m}^3/\text{min}$ (1 mark)

✓ correct capacity of cut Y

(b) Is the maximum flow through the system of ducts from A to H equal to the largest of the capacities you found in part (a)? Justify your answer. (1 mark)

No, max flow cannot be more than the smaller cut of $65 \text{ m}^3/\text{min}$

✓ states no with reasonable justification

(c) State the maximum possible flow along the path ADEFH. (1 mark)

$4 \text{ m}^3/\text{min}$

✓ correct flow

(d) Determine the maximum flow through the system of ducts from A to H.

(3 marks)

5 marks for 15, 10, 10, 10, 4, 6, 10

$$ABEH = 15$$

$$ABEFH = 5$$

$$ABFH = 10$$

$$ACFH = 10$$

$$ADEFH = 4$$

$$ADFH = 6$$

$$ADGH = 10$$

$$\text{Max flow} = 60 \text{ m}^3/\text{min}$$

or Cuts

$$\text{Cut to left of EFH} = 60$$

$$\text{Cut through source} = 65$$

$$\text{Cut to right BCD} = 75$$

$$\text{Max flow} = 60 \text{ m}^3/\text{min}$$

✓ value of new cut

✓ values of 2 more

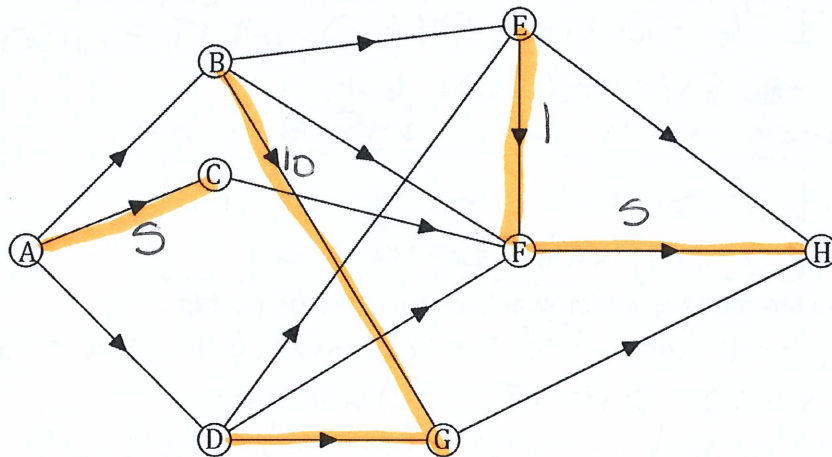
✓ states correct max flow

✓ correct flow along at least 1 path

✓ correct flows along at least 2 more paths

✓ states correct max flow

(e) To achieve the maximum flow determined in part (d), not all ducts will be used to their maximum capacity. Clearly label all such ducts with their spare capacity on the copy of the network below. (2 marks)



See diagram

Note: $EF(1) + FH(5)$ may be replaced with $EH(1) + FH(4)$

✓ correctly identifies + labels 2 or more
✓ correctly identifies + labels all ducts

Question 19

(8 marks)

Guy currently has a nil balance in his savings account that earns 5.04% per annum, compounded monthly. He plans to fund a 12-month round-the-world trip by making deposits of \$1150 into the account at the end of each month.

- (a) Determine the balance of his account just after his 6th deposit. (2 marks)

$$N = 6, I\% = 5.04, PV = 0, PMT = -1150, PY = 12$$

$$CY = 12 \rightarrow FV = 6972.86$$

Balance will be \$6972.86

✓ sets at least 5 correct values in Fin Calc
✓ correct balance

As soon as his balance first exceeds \$48 000, Guy will stop making deposits and head off on a 12 month trip. Because regular payments are no longer made into the account, the interest rate will decrease to 4.56% per annum. Guy will make the same withdrawal at the end of each month, so that after the 12th one his savings account balance will have fallen to \$5000.

- (b) Determine the balance of his account when he stops making deposits. (2 marks)

$$I\% = 5.04, PV = 0, PMT = -1150, FV = 48000$$

$$PY = 12, CY = 12 \rightarrow N = 38.5$$

$$N = 39, I\% = 5.04, PV = 0, PMT = -1150, PY = 12$$

$$CY = 12 \rightarrow FV = 48621.64$$

Balance will be \$48 621.64

✓ correct value of N
✓ correct balance

- (c) Determine the monthly withdrawal Guy will make on his trip. (2 marks)

$$N = 12, I\% = 4.56, PV = -48621.64, FV = 5000$$

$$PY = 12, CY = 12 \rightarrow PMT = 3744.55$$

Withdrawal must be \$3744.55

✓ sets at least 5 correct values in Fin Calc
✓ correct amount of withdrawal

- (d) Determine the total amount of interest that his savings account earned from the time Guy started saving for his trip until the end of the trip. (2 marks)

Let I be the interest, $39 \times 1150 - 12 \times 3744.55 + I = 5000$

$$I = \$5084.60$$

Using Amortization will result in $3771.64 + 1312.95 = 5084.59$

✓ indicates correct method
✓ correct total interest

End of questions

Supplementary page

Question number: _____

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